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Generative AI and the augmentation of information practices in knowledge work

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ABSTRACT

In this article, we explore the integration of generative AI systems into knowledge work, focusing on how these systems augment key information practices. Drawing on a mixed-method approach that includes digital diary studies and interviews, our human-centred exploration of interactions between knowledge workers and generative AI identifies the roles these systems play in complementing workers' contributions to information representation, generation, evaluation, explanation, and communication. We find that generative AI systems are increasingly taking on augmentative roles such as synthesiser, organiser, brainstormer, reviewer, interpreter, and clarifier. Our findings also highlight the active role of knowledge workers, revealing a dynamic human – AI partnership that improves the efficiency of knowledge work, fosters creativity, and supports decision-making processes. This partnership not only leverages generative AI's capabilities to augment information practices but also amplifies and draws upon distinct human expertise in critiquing, integrating, contextualising, communicating, and individualising AI-generated outputs.

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Generative AI; knowledge work; AI affordances; information practices; sensemaking; human-AI collaboration; division of work between humans and AI; AI literacy

1. Introduction

The recent advent of generative AI has captured the attention of the general public, academia, and innovators across the board. These systems, primarily powered by deep learning technologies, are capable of producing original content across various modalities, including text, images, videos, audio, and computer code (Ellingrud et al. 2023; Fui-Hoon Nah et al. 2023). While generative AI encompasses more than just text generation, this article places a particular emphasis on large language models (LLMs), such as ChatGPT, which are seen as situated at the forefront of generative AI's ongoing development. However, when we refer to 'generative AI systems' in this article, we acknowledge the limitation that not all of these systems are focused on language. One of the primary strengths of generative AI systems is their ability to transform information activities and knowledge processes (Benbya, Strich, and Tamm 2024; Feuerriegel et al. 2024), thereby enhancing the creation, modification and analysis of information to meet the diverse needs of various user bases (Choi, Zhang, and Stivilia 2023; Hirvonen et al. 2024). For instance, with respect to text, recent models can handle open-domain, multi-turn dialogues while executing a diverse set of natural language processing (NLP) tasks such as text categorisation, retrieval, summarisation, and more. These capabilities create clear

implications for the transformation of knowledge work (Jiang, Karanasios, and Breidbach 2024; Söllner et al. 2025).

The implications of these technologies for the workforce and the potential to replace knowledge workers have been subjects of considerable debate and scholarly discussion. This debate has contributed to the ongoing dialogue about the role of AI within the work context, which has both alarmed and excited organisations and individuals for several years now (Ellingrud et al. 2023; Jarrahi, Möhlmann, and Lee 2023). Though not without its limitations, these systems offer unprecedented capabilities compared to the previous generation of systems (Hui, Reshef, and Zhou 2024; Jaffe et al. 2024) due characteristics such as generativity, self-learning ability (Faraj, Pachidi, and Sayegh 2018), naturalistic interactional paradigm, and more. For instance, new genAI systems can implicitly infer instructions from examples and accurately capture some forms of tacit knowledge that were previously thought to be attainable by workers only through their lived experiences (Brynjolfsson, Li, and Raymond 2025). These characteristics, often emergent, foreshadow the potential for genAI systems to pervade ever more tacit aspects of knowledge work that were previously deemed immutable to information system automation (Benbya, Strich, and Tamm 2024; Devanny, Dylan, and Grossfeld 2023).

While recent research suggests that generative AI can enhance knowledge workers' productivity by boosting the efficiency of routine information work tasks, much remains unknown given the nascent and fast evolving nature of this phenomenon. To better understand and evaluate the workplace implications of AI, recent analyses tend to gravitate towards two extremes. On one hand, there's a focus on large-scale economic assessments of tasks and professions, primarily concerned with the extent of 'exposure' to AI-based automation and potential job displacement (e.g. Eloundou et al. 2024; Felten, Raj, and Seamans 2023; Webb 2020; Zarif-honarvar 2024). On the other hand, there are more lab studies or user surveys, comparing human and AI performance in specific knowledge-intensive tasks, often isolating these tasks for micro-comparisons (e.g. Cam-bon et al. 2023; Gilardi, Alizadeh, and Kubli 2023; Nas-cimento, Alencar, and Cowan 2023; Noy and Zhang 2023). Both approaches, however, share important limitations. Firstly, they tend to overlook the unique contextual aspects of work, failing to capture the on-the-ground realities of how knowledge workers engage with generative AI systems on a day-to-day basis. Consequently, there exists a gap in terms of a grounded understanding of how these professionals incorporate and adapt AI uses into their work environments. This deficiency undermines the recognition of knowledge workers' agency in making sense of AI and adjusting to its presence (Lindley and Wilkins 2023). Secondly, both macro and micro analyses implicitly promote a narrative of automation and competition, where humans and AI systems are seen as adversaries vying for supremacy. Such a perspective neglects to consider AI's technological feasibility when coupled with considerations for organisational realities such as ethical, legal, and policy implications (Frey and Osborne 2023).

Therefore, less attention has been paid to how AI systems play out over time for knowledge workers for whom daily information-centric activities (Hirvonen et al. 2024; Ramaul, Ritala, and Ruokonen 2024) form the foundational core of the broader set of processes and practices (Brachman et al. 2024; Jarrahi et al. 2023). Addressing the set of limitations outlined above, our study grounds AI and its functionalities in the context of key information practices, providing an understanding of how generative AI systems are being integrated into daily work. Using this approach, rather than focusing on the replacement narrative of AI, we specifically examine how these systems complement knowledge workers' roles in completing these information practices.

Inspired by Human-Centred Artificial Intelligence (HCAI) perspective, our research adopts an approach

that treats AI not as an autonomous driver of work effectiveness that displaces knowledge workers, but rather as a chance to enhance human abilities and performance based on their own needs, practices, and aspirations (Fui-Hoon Nah et al. 2023; Schmager, Pappas, and Vassilakopoulou 2023). This viewpoint supports a collaborative relationship between humans and AI systems, promoting the concept that, together, they can accomplish more than either can individually.

Our investigation is guided by the following research questions:

- How may generative AI systems augment the information practices of knowledge workers?
- In what ways do knowledge workers play a complementary role alongside these AI systems?
- How does the emerging division of labour between humans and AI manifest in the uses of these systems in daily information practices?

In addressing these questions, our aim in this research is to enhance the nuanced understanding of how generative AI is transforming the landscape of knowledge work and the collaborative dynamics between humans and AI systems.

In the remainder of this article, we first review the context of relevant literature with a focus on the emerging literature on generative AI and knowledge work, the human-centred AI perspective, and the role of information practices in shaping AI adoption. In our review of the literature, we specifically draw attention to information practices perspective that examine how knowledge workers engage with AI-generated information. The methodology section then details the research design, data collection, and analytical approach. In the findings section we present key insights into how research participants incorporate, evaluate, and adapt AI systems. Finally, in the discussion and conclusion we reflect on the broader implications of AI augmentation in knowledge work, highlighting areas for future research and practical considerations.

2. Literature review

This literature review section considers three interconnected themes that shape the foundation of our research approach: how generative AI is shifting from automation to augmentation in professional tasks, the human-centred AI perspective that views AI as assistive rather than autonomous entity, and information practices as the context within which AI shapes knowledge workers workflows. Together, these themes provide the foundation for contextualising how knowledge

workers may adopt AI in their daily information practices.

2.1. Generative AI in knowledge work

Recent studies on generative AI highlight its potential to both automate and augment human labour in knowledge work. Since the public release of systems like ChatGPT in late 2022, research on the potential consequences of these systems in the work domains has rapidly accelerated (Ayinde et al. 2023; Feuerriegel et al. 2024). The multi-disciplinary research has evolved from early controlled experiments to more contextually-rich studies of organisations and work practices.

Initial research focused primarily on experimental settings, evaluating generative AI on discrete tasks such as writing and coding (Cambon et al. 2023; Gilardi, Alizadeh, and Kubli 2023). This work often focused on the unique capabilities of these systems in interacting with humans or performing tasks that were previously deemed the exclusive realm of humans in controlled experimental settings. For example, Park et al. (2023) suggest that generative AI agents, when equipped with a comprehensive architecture, can provide believable simulations of human behaviour in interactive environments, while Bommasani et al. (2021) showcased unique generative AI capabilities in summarising complex research papers with accuracy comparable to domain experts.

Empirical studies have also documented significant efficiency gains when people started to use generative AI in work-related contexts (Söllner et al. 2025). Noy and Zhang (2023) found that systems like ChatGPT enabled users to complete writing tasks in 37% less time. Likewise, Spatharioti et al. (2023) demonstrated that users solved decision-making problems twice as fast using AI-based search compared to traditional methods. AI is therefore seen as a productivity booster for knowledge workers particularly when the tasks fall within its 'jagged technological frontier,' while AI use for tasks outside this frontier may result in decreased performance (Dell'Acqua et al. 2023). This research concludes that the positive productivity effects observed in lab and experimental settings are likely to manifest in real-world work environments (Jaffe et al. 2024).

While research has documented the overarching impact of these AI systems, it has also highlighted their uneven impact across different roles and skill levels (Hui, Reshef, and Zhou 2024). Brynjolfsson, Li, and Raymond (2025) found that generative AI significantly improved task resolution rates for low-skilled customer service workers (by 34%), while its effect on the productivity of experienced workers was not as visible.

This often means less experienced workers see improvements in both speed and quality of their output because of generative AI adoption, while highly experienced counterparts either see small gains in speed or even small declines in quality of performed tasks (Maslej et al. 2023). These findings highlight the complex integration of AI into knowledge work, reinforcing that AI adoption is not simply a process driven by AI and key technological capabilities, rather it is a process of augmentation contingent on human-AI synergies in dynamic work environments.

Despite the promise of AI-driven augmentation of work, the current research landscape remains fragmented. Studies of AI at work either largely focus on large-scale macroeconomic projections of AI-induced labour shifts (Acemoglu and Restrepo 2020; e.g. Eloundou et al. 2024; Zarifhonarvar 2024) or micro-level analysis of task performance in controlled experiments (e.g. Gilardi, Alizadeh, and Kubli 2023; Noy and Zhang 2023). Both approaches are valuable but tend to obscure the everyday realities of AI adoption in knowledge-intensive work, in which workers exercise discretion in integrating AI into their workflows and are subject to contextual nuance (Hirvonen et al. 2024; Lindley and Wilkins 2023). Less attention has therefore been given to how knowledge workers make sense of AI outputs, and incorporate AI-generated information into their practices. As a result, more empirical research is needed to examine how generative AI embeds itself in real-world information practices that undergird knowledge work, how knowledge workers negotiate their role in their work, and how human-AI collaboration evolves over time.

This study contributes to this gap by applying an information practices perspective to generative AI in knowledge work. Rather than framing AI as an autonomous entity replacing human labour, we examine how knowledge workers actively incorporate AI-generated outputs into information representation, evaluation, and communication. By centring human agency in AI adoption, this approach aligns with human-centred perspectives on AI.

2.2. Human-Centred perspective towards human-AI partnership

The human-centred approach to artificial intelligence (HCAI) emphasises designing AI systems that augment rather than replace human abilities by keeping human judgment, values, and agency central in AI-augmented workflows (Capel and Brereton 2023; Shneiderman 2022). In contrast to automation-centred paradigms that prioritise efficiency and autonomy through AI,

HCAI envisions AI as a tool that extends human cognition, creativity, and decision-making capabilities while preserving meaningful human control (Schmager, Pappas, and Vassilakopoulou 2025). HCAI perspective is particularly relevant as generative AI is increasingly integrated into both daily life and work, impacting how we interact with and make use of AI-generated information.

A key principle of HCAI is human control – ensuring that AI remains a system assisting humans rather than an autonomous decision-maker (Shneiderman 2020). In high-stakes domains such as healthcare or criminal justice, AI-driven decision support systems can provide recommendations to human decision makers, but humans eventually retain final authority over decisions (Xu et al. 2023). This approach mitigates the risks associated with over-reliance on AI, preventing errors that arise when users assume AI-generated outputs to be infallible (Strauß 2021). By keeping humans in the loop, HCAI sees AI as a resource for augmenting human expertise rather than dictating outcomes and designing end-to-end autonomous workflows.

Transparency and accountability also play crucial roles in HCAI. One of the principles of HCAI is to develop AI systems with interpretable outputs so that users can understand how AI makes decisions and critically evaluate AI-generated outputs (Ehsan and Riedl 2020). Without transparency, AI systems risk becoming opaque, eroding trust in human-AI interactions, and making it difficult for users to work effectively alongside the system and spot errors and biased performance (Chen et al. 2023). This principle is particularly relevant in knowledge work, where the worker must be able to assess, verify, and contextualise AI-generated content. Accountability mechanisms, such as keeping the human in supervisory roles, ensure that AI remains a complement to human expertise and context rather than an unchecked authority (Jarrahi and Ahalt 2024).

A defining aspect of HCAI approach is the recognition that AI's role is to **augment** human cognition and role, not replace it (Schmager, Pappas, and Vassilakopoulou 2025). While AI can process vast amounts of data, identify patterns, and generate new content at an unprecedented scale, it may lack some aspects of tacit knowledge, contextual awareness, and critical reasoning that define human expertise (Felin and Holweg 2024). Therefore, AI-generated outputs still require humans to be interpreted, refined, and integrated into broader knowledge ecosystems (Hirvonen et al. 2024). Humans still play a critical role in validating, adapting, and making sense of AI-generated insights particularly when issues such as hallucination continue to remain a feature rather than a bug of generative AI system (Sison et al.

2024). This shift from automation to augmentation serves as the core of HCAI, reframing AI as a **collaborative tool** that must enhance human capabilities to enable more efficient, informed, and creative work processes (Topali et al. 2024).

In the context of generative AI and knowledge work, the argumentation model, espoused by HCAI, means the effectiveness of AI systems is contingent on human intervention – workers must assess AI-generated content for relevance, accuracy, and coherence, ensuring it complies with context-specific standards and domain knowledge (Schmager, Pappas, and Vassilakopoulou 2025). Similarly, in decision-making contexts, HCAI argues that knowledge work is augmented most when AI tools are designed and deployed to facilitate interactive collaboration, where users iteratively refine AI outputs rather than passively accepting or integrating them in their work (Shneiderman 2022). Such a human – AI partnership approach is exemplified in concepts like 'hybrid intelligence' or 'centaur' work strategies, through which humans and AI complement one another and jointly contribute to outcomes (Jarrahi, Lutz, and Newlands 2022; Mollick 2024).

Despite the promise of AI augmentation, the reality of integrating generative AI into knowledge work remains complex. To achieve the goals articulated through the HCAI perspective, of designing future systems that are human-first rather than autonomy-first, there needs to be a more contextual understanding of how workers may partner with AI systems on the ground. Therefore, in this study, we contribute to this perspective by examining how knowledge workers over-time engage with generative AI specifically through the lens of information practices.

2.3. Information practices in the context of knowledge work

One effective approach to capturing the actual roles technological systems play in supporting work involves observing how humans use these systems in specific information-related activities (Lloyd 2010). This usage highlights how certain technologies enable or restrict different types of actions (Jones and Norris 2005). Therefore, to comprehensively understand the benefits and role of tools like generative AI systems in augmenting knowledge work, one can focus on the information practices that serve as the foundation of knowledge work (Hirvonen et al. 2024; Ramaul, Ritala, and Ruokonen 2024; Zhong, Han, and Hansen 2023). The term 'information practice' describes actions that are shaped by social, cultural, and historical contexts and facilitated through tangible means to identify, create, search for,

manage, use, and share information. These practices are recurrent, grounded in physical reality, and specific to particular situations (Zhong, Han, and Hansen 2023), reflecting how workers engage with information in their daily tasks (Orlikowski 2002).

Information practices involve complex interactions among individuals, technology, and information (Cox 2012). They encompass the myriad of ways in which knowledge workers navigate, manipulate, and construct information landscapes to achieve their professional objectives (Lindley and Wilkins 2023). Deeply embedded in the fabric of organisational life, these activities both influence and are influenced by the broader sociotechnical environment (Heyder, Passlack, and Posegga 2023). Consequently, examining information practices provides valuable insights into the dynamic interplay between human agency, technological affordances, and the information ecosystem (Cox 2012). As such, in this article, ‘information practice’ serves as a framework for understanding how the use, benefits, and limitations of generative AI systems are applied within the context of knowledge work.

Our research identifies the following – information representation, information generation, information evaluation, information explanation, and information communication – as the most relevant information practices that are transformed or augmented by the adoption of generative AI systems.

2.3.1. Information representation

Information representation generally refers to the methods and processes by which existing information is organised, structured, and presented to the user (T. Lee, Lee-Geiller, and Lee 2020). This encompasses how information is modelled, visualised, and made accessible to users, influencing how effectively they can interpret and integrate it into their daily lives. Transparent and coherent information representation is understood as a crucial practice for enhancing user understanding and accessibility, and it has been thoroughly examined in the previous literature (e.g. Bates 2005; Dourish and Mazmanian 2011).

2.3.2. Information generation

Information generation involves the creation of new information or insights, relying on not only synthesis of existing information but also creative and innovative approaches (Hirvonen et al. 2024). This practice is crucial for organisational learning and knowledge creation. Specifically, the capacity to generate new and contextually relevant information is critical for problem-solving and developing new ideas and concepts (Zhang et al. 2023). This, in turn, emphasises its importance

in various contexts, including knowledge management (Jarrahi et al. 2023).

2.3.3. Information evaluation

Information evaluation entails the critical assessment of information to determine its accuracy, relevance, and reliability (Zhong, Han, and Hansen 2023). Ensuring the quality and validity of information used for decision-making continues to be a key practice undergirding knowledge work (Widén, Ahmad, and Huvila 2023). The criteria and methodologies used in information evaluation assess the credibility of information sources by emphasising rigorous standards to evaluate the authenticity and reliability of information (Rieh and Danielson 2007).

2.3.4. Information explanation

Information explanation focuses on clarifying, elucidating, and making information comprehensible to diverse audiences (Savolainen 1995). This information practice involves breaking down complex concepts, translating technical jargon, and providing context. The role of explanation in enhancing user understanding and knowledge transfer is critical as it ensures that users can grasp the underlying meaning, context, and significance of the information presented (Tufte 1991).

2.3.5. Information communication

Information communication involves the effective transmission and sharing of information among individuals and in communities of practice (Bronstein and Solomon 2021; Robson and Robinson 2013). This practice often includes selecting the appropriate medium, tone, and style to convey information clearly and persuasively to specific groups of people (Byrne and LeMay 2006). Effective communication is central in knowledge management by impacting how information is received and consumed by its intended audience (Davenport and Prusak 1998).

3. Methods

To examine how knowledge workers interact with generative AI systems in real-world scenarios, our data collection approach comprised two main elements. Each participant undergoes: (1) a 10-day digital diary study, followed by (2) a semi-structured interview with members of the research team.

We employed a cohort-based model to facilitate participants’ engagement in both the diary study and the semi-structured interviews (see Table 1 for more details about the participants). That is, groups of participants began the diary study on the same day, wrote entries

Table 1. Participants demographic and occupation information.

	Age	Gender	Role	AI tool(s)
P1*	30–39	Female	Professor	ChatGPT
P2*	<30	Male	Researcher	ChatGPT
P3*	50–59	Female	English language specialist	ChatGPT
P4	<30	Female	Clinical research coordinator	ChatGPT
P5*	<30	Female	UX Designer	ChatGPT
P6	<30	Female	Customer service agent	ChatGPT, Bard, Bing
P7	30–39	Male	Healthcare professional	ChatGPT
P8	<30	Female	Healthcare professional	ChatGPT
P9	30–39	Female	Project manager	ChatGPT
P10	<30	Male	Solutions engineer	ChatGPT, Github Copilot
P11	<30	Female	Sales manager	ChatGPT
P12	30–39	Female	Researcher	ChatGPT
P13	40–49	N/A	Program director	ChatGPT
P14	<30	Male	Software developer	ChatGPT
P15	<30	N/A	Data science analyst	ChatGPT
P16	<30	Male	Researcher	ChatGPT
P18	<30	Female	IT consulting	ChatGPT
P19	30–39	Female	College counsellor	ChatGPT
P20	60 and over	Female	Independent consultant	ChatGPT, Bard, Bing
P21	60 and over	Male	Professor	ChatGPT, Bard, Claude
P22	30–39	Male	Healthcare consultant	ChatGPT
P26	N/A	N/A	Cybersecurity manager	ChatGPT
P27	<30	Female	Researcher	ChatGPT
P28	50–59	Male	Digital media program manager	ChatGPT, Notion AI, Adobe Firefly, Miro AI, Quillbot, MidJourney

* These participants completed the interviews but not the diary study portions of the study.

for at least 10 days, and then completed the individual semi-structured interviews. Once a group of participants completed the end-to-end process, the research team brought in a new cohort of participants. While an existing group was taking part in the research process, a new group of participants was recruited and identified for the next round of diary and interview data collection. With each new cohort enrolled, we sought to iteratively fine-tune the data collection process based on feedback from prior rounds.

Participants were recruited through various channels, mainly via UNC's ResearchForMe web portal, linked to an approved IRB. This portal enabled researchers to detail their study, target participants, provide compensation information, and offer other relevant details, allowing site visitors to express interest and complete a preliminary survey. This survey collected data on the respondents' use of generative AI, their occupation, and contact details if they qualified for the study. It also informed them about the study's methodology, including diary and interview stages. Participants willing to engage in both components provided written

consent. Recruitment efforts also extended to social media platforms, like LinkedIn and Reddit, and information professional listservs.

The collection process involved participants documenting their use of generative AI at work for 10 days within a 3-week diary study. First, a welcome email outlined the study period and shared insights from previous cohorts. Participants received daily morning and afternoon reminder emails, along with a survey link to record their interactions. The morning reminder prompted documentation of generative AI tool use, and the afternoon reminder prompted submission of these notes. Participant privacy was prioritised, ensuring they shared only what they were comfortable with (see Figures 1 and 2 for examples of entries and attachments participants submitted). Additionally, the research team conducted weekly check-ins to encourage quality entries and provided regular email updates to maintain engagement, including information about upcoming interviews.

The goal of the diary study was not only to capture in situ (in its original place of occurrence) how knowledge workers were using generative AI tools but also, through the process of documenting their usage, provided participants with the time and space to reflect on the actual utilities and limitations of generative AI use in their specific work contexts (Jarrahi et al. 2021). In terms of the reported usage, while OpenAI's ChatGPT was the most popular system, participants were not limited to specific generative AI tools. Thus, some also described their usage of Google Gemini, Microsoft Bing, and Anthropic's Claude.

Before the conclusion of the diary study for a cohort, the research team would reach out to the existing cohort and provide reminders to begin scheduling individual semi-structured interviews. Because each participant might finish their 10-day diary at different times, the semi-structured interviews were staggered. Due to the length of the data collection period (approximately 4 weeks total), there were some drop-offs in the process. However, for those who were able to complete the interview, the interview process aimed to (1) synthesise and echo back to participants the interactions, challenges, insights they shared in their diary entries, (2) probe and further unpack any elements of the diary entries that were unclear, and (3) broaden the conversation to elicit additional reflections and commentaries participants might have regarding generative AI systems. Prior to each interview, the research team reviewed and thematised all diary entries submitted by the participant and tailored the interview protocol accordingly.

The interviews were conducted via Zoom. To begin, participants were asked to provide background

Q2. Date and time of the session: February 27, 2024

Q3. In what ways did ChatGPT/other LLMs help or support your work today? Please provide details about the tasks or problems ChatGPT/others assisted you with.

Today I worked on reviewing material for a resource I am working on. The resource contains a lot of information about co-designing with teens. The resource is in Notion, and I am using Notion AI to help me with the writing. I write the description, then select Improve Writing. I still have to change the text, because at times the tone is very wordy, but it helps me organize my thoughts and also identifies the areas in the writing where I need to improve. I do believe that AI, if used responsibly, is a very useful tool. Using it responsibly includes not just taking the results at face value but rather reading them carefully and modifying them when necessary.

Figure 1. Sample diary entry describing one participant's information practice.

information regarding their work context. The interview was then structured into four parts. In part one, we investigated participants' information practices as noted in the diaries, along with any other practices not captured. We further probed about the usage patterns, use cases, key benefits of generative AI tools, and how the integration of generative AI impacted the use of other information tools that participants previously relied on. In part two, the interview moved towards a discussion about the issues and limitations surrounding the uses of generative AI systems such as managing information accuracy and other ethical considerations of generative AI tool use. Part three extended the conversation towards how participants learned to use the systems, and whether strategies and practices have been developed over time. Finally, part four of the interviews shifted the discussion towards considerations regarding implications of these tools on participants' skill set and job. After completing the interview, participants were asked to fill out an outtake survey to provide basic demographic information, as well as additional feedback about their experience to further improve the data collection process for the following cohort. Participants were compensated with \$60 for completing the diary study and the interview. Overall, three pilot interviews were conducted to refine the initial protocol. The durations of the interviews varied from 30 to 90 min. This variability arises in part because the interviews were highly tailored to the diary entries that participants submitted. The more in-depth the entries were for a participant, the more tailored and specific the interviews became.

We began data collection in May 2023, following approval from the university's Institutional Review Board (IRB) for human subjects research. Although

data collection continued through May 2025, resulting in 9 completed cohorts, this article includes only the first five cohorts in its analysis. Participant backgrounds included academic researchers and administrators (e.g. professor, doctoral student, clinical research coordinator, programme director), IT workers (e.g. developer, UX designer, project manager), healthcare professionals, education specialists (e.g. college counselors), and business consultants. For this paper, the analysis is focused on the information practices of participants from the first five cohorts of the study.

In our analysis of the data, we utilised a two-stage qualitative coding process (Charmaz 2008). We started with open coding to pinpoint essential information practices and the related roles of generative AI in supporting those practices, as revealed through diary entries and interviews. This step led to axial coding, which linked these identified categories to examine the dynamics between the information practices and generative AI functions. This systematic approach revealed the complex ways in which knowledge workers incorporate these technologies into their work with an emphasis on the strategies they use to benefit from AI tools and the obstacles they encounter. Our iterative theme refinement process resulted in a comprehensive examination of the changing interactions between professionals and generative AI systems. We identified crucial roles played by generative AI that influence information practices, which offer valuable insights for both future research and practical implementations in the domain (see Table 2).

4. Findings

In this section, we present our findings on key information practices, the argumentative roles of

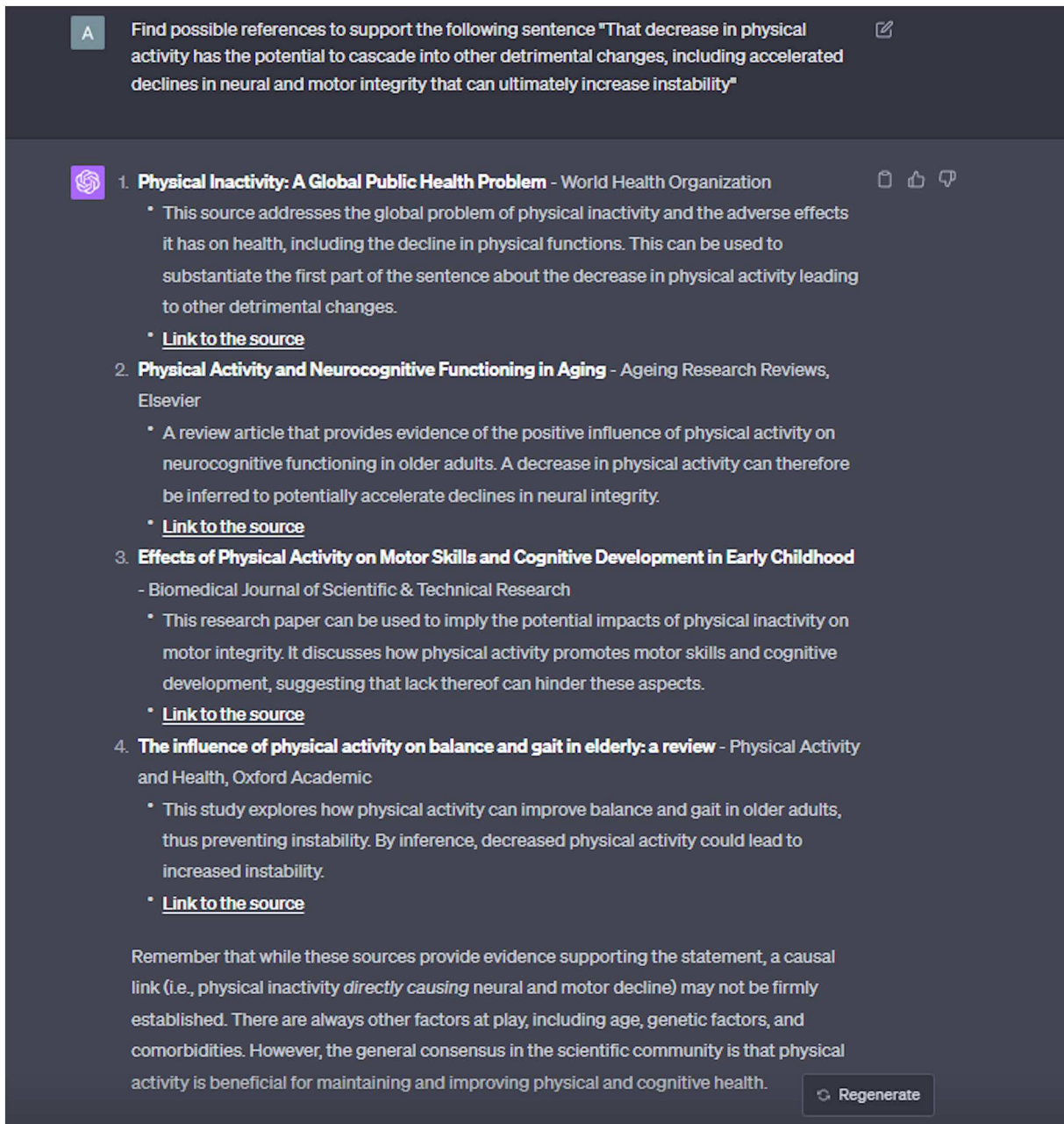


Figure 2. Sample chat log shared by a participant during the diary study.

generative AI systems in scaffolding those practices, and critical roles played by knowledge workers (see Figure 3).

First, we leverage these practices to contextualise the role of generative AI within the realm of knowledge work (see Table 2). It is important to note that these functions are intrinsically linked and overlapping, rather than being mutually exclusive, and may be combined in each interaction with the system. Then, we describe the roles that knowledge workers engage in, across information practice categories, to tie together the component outputs produced by generative AI.

4.1. Information representation

Information representation emphasises effective methods for retrieving, synthesising, organising, and presenting existing information scattered across various resources in new forms, such as prose, tables, slides, or lists. Generative AI's role in this context can thus be crucially understood as synthesisers, organisers, and retrievers of information. For all of these roles, generative AI systems rely on existing information available on the Internet or information provided by the user (e.g. meeting notes) to respond to a prompt. Therefore, the

Table 2. Information practices and corresponding augmentative role of generative AI systems.

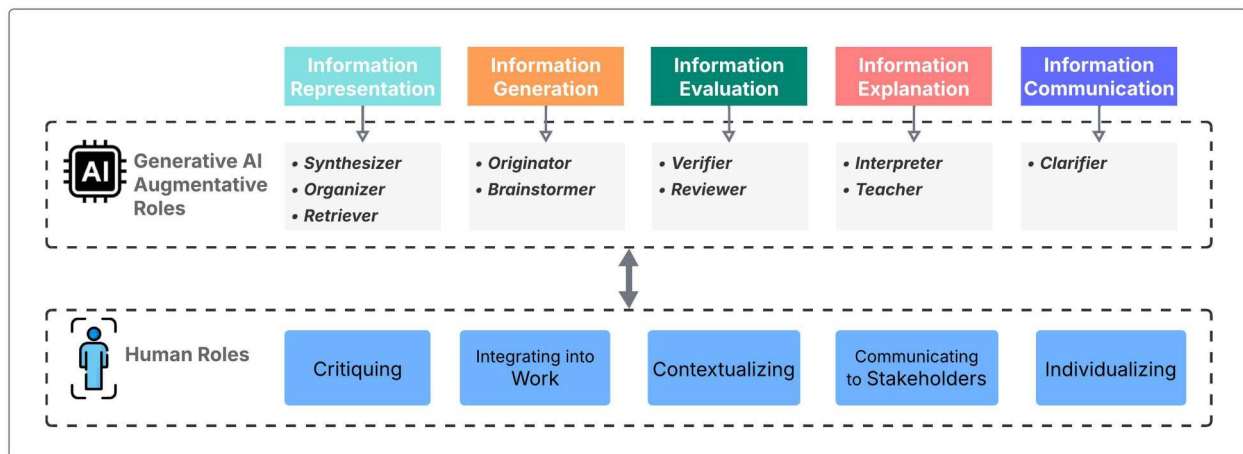
Information practice	Definition	The augmentative role of generative AI
Information representation	Focuses on retrieving, synthesising, organising, and presenting existing but scattered information in new, coherent forms.	Synthesiser: Integrating and recombining information from diverse sources to produce coherent and relevant responses, focusing on synthesis and summarisation. Organiser: Systematically categorising and structuring diverse information into coherent and orderly formats, making it easier for users to navigate through various topics, ideas, and solutions. Retrievers: Providing direct and efficient response delivery to queries (overlapping with that of traditional search engines like Google).
Information generation	Focuses on generating entirely new content or innovative insights that are contextually relevant to move beyond mere representation by adding unique value or perspectives.	Originator: Initiating the foundational basis for various information-related tasks by creatively generating initial concepts, frameworks, or datasets to effectively set the stage for further development or analysis. Brainstormer: Serving as a platform off which to bounce ideas in an iterative manner, thereby acting as creative catalysts in the brainstorming process.
Information evaluation	Focuses on examining and validating information to ensure its accuracy, completeness, reliability, and freedom from errors.	Verifier: Analysing and assessing content for accuracy, relevance, and coherence to ensure the information presented aligns with the context and requirements. Reviewer: Proofreading and editing text to ensure clarity and grammatical accuracy, thereby enhancing the overall quality and structure of written content.
Information explanation	Focuses on clarifying, elucidating, and providing meaning to information.	Interpreter: Translating and clarifying specific information, such as language or technical jargon, as well as facilitating a deeper understanding of the diverse interpretations and nuances of that information. Teacher: Breaking down complex concepts, providing detailed explanations, and unpacking ideas. In this capacity, AI assists in imparting a holistic understanding more broadly, which sets it apart from mere information synthesis or interpreting specific information.
Information communication	Focuses on the effective transmission and sharing of information and ideas with other parties.	Clarifier: Clarifying information to effectively communicate with diverse audiences and determining the most suitable tone by analysing language subtleties and contextual nuances to provide recommendations that enhance message clarity, persuasiveness, and audience-specific tailoring.

common thread among these functionalities is presenting existing information in a useful and often new way to more effectively meet the user's information needs.

4.1.1. Synthesiser

A key feature of these systems is the integration and recombination of diverse pieces of information, often

dispersed across various sources, to create coherent and relevant responses. At its core, this role of AI focuses on bringing things together. Specifically, participants emphasise these systems' abilities to synthesise and summarise information by condensing large volumes of data into brief, focused, and comprehensible information. Information synthesis capacities of

**Figure 3.** Information practices and augmentative roles of generative AI systems and humans.

generative AI highlights essential points and main ideas to help participants more easily understand the core content of extensive informational materials. Examples of extensive informational materials noted in the diary entries included primarily text-heavy documents like extended reports, interview transcripts, programming files, and academic literature.

P18 pointed out that synthesis was one of the key value additions of ChatGPT due to its time saving ability. Specifically, P18 mentioned, *'It [ChatGPT] removes a step, almost, because you don't have to go through different articles, and it'll just give you a response.'* In traditional search scenarios, synthesis largely depends on the human's ability to assess the credibility of a source and find the information of interest within reputable sources. This assessment is particularly complicated for tasks that go beyond the simple retrieval of facts into the realm of opinions and subjective topics. ChatGPT offers strong synthesis capabilities to address these gray areas. Specifically, ChatGPT often spares users the need to sift through numerous search results and piece them together to address a specific question by, instead, offering a direct answer to the question in prose.

ChatGPT users who need to rely on others' domain expertise gain particular value from the synthesis capabilities because it allows them to more quickly put new information into practice. For example, P19 discussed how coaching volleyball and creating a practice plan could require an excessive amount of time and energy. ChatGPT can assist in synthesising proven, successful volleyball practice plans based on the experiences of existing volleyball coaches.

4.1.2. Organiser

This relates to the key capacity of emerging AI systems in systematically categorising and structuring diverse information into coherent, orderly formats to make it easier for users to navigate through various topics, ideas, and solutions.

For instance, P13's use of ChatGPT as an organiser included tasks such as cleaning up meeting notes or organising meeting agenda. These organising capacities of the systems are also utilised to understand a large task or goal before breaking this big idea down into *'actionable steps or small steps,'* as P18 puts it. The role of the system as organiser is key here to ensure that all of the newly produced steps are cogently organised and contribute to overall coherence.

ChatGPT also facilitates organising across multiple formats. P28 exported content from Canva or Google Slides and then uploaded it into ChatGPT to organise various formats and disparate materials into presentation slides. ChatGPT's capabilities can be equally

valuable for organising other types of data beyond the previous examples. P10, for instance, increased efficiency and saved time by using the system to arrange tabular data into various formats such as CSV, rather than writing scripts or employing other tools.

4.1.3. Retriever

Generative AI systems also provide direct and precise response delivery to participants' queries; this role largely overlaps with that of traditional search engines like Google. In contrast to traditional Google search, which requires the extra step of processing relevant results, ChatGPT generates a direct answer in prose. As SEO-optimised, AI-generated, and advertiser-funded content floats to the top of search results, thereby rendering search engines like Google increasingly biased towards advertising, ChatGPT can emerge as a more efficient alternative. P10, for example, explained that ChatGPT was helpful in retrieving information because the participant would no longer have to *'wade through a bunch of ads'* as they previously had with Google.

This functionality is particularly relevant for questions that rely on retrieving information that is extremely likely to exist, but not at hand. For instance, P26, a cybersecurity consultant, used ChatGPT to inquire about a specific business's operating hours. Rather than having P26 visit the business's website, check their Facebook page, or conduct a manual search, ChatGPT consulted information from Internet sources and provided a functional answer. Notably, given their highly technical background, P26 had complete trust in ChatGPT's response because P26 felt confident in understanding how ChatGPT worked. Specifically, P26 trusted that the language model would provide a correct answer for a query like this example so long as the source information was correct. In other words, as long as the business had correctly updated any changes to their hours online, P26 believed that ChatGPT would be able to scrape this information from the Web and generate an accurate response.

4.2. Information generation

Information generation, as a practice, involves producing novel information and, more specifically, innovative ideas that are contextually relevant. While there is some overlap with information representation, the key distinction lies in the nature of the content produced: information representation focuses on repackaging existing data, perhaps in creative ways, whereas information generation entails creating entirely new content

or insights that go beyond mere representation to add unique value or perspectives to the discussion.

4.2.1. Originator

AI as the originator corresponds to the role these systems play in establishing a foundation for various information-related tasks. Participants described the system doing so by generating initial concepts, frameworks, or datasets. This foundation would then effectively set the stage for further development or analysis by providing users with a starter version of the data or document from which to work.

ChatGPT is widely recognised for its ability to generate initial drafts for a variety of content commonly used by workers. P20 highlighted this feature, explaining how ChatGPT served as a 'baseline' that could be easily tweaked with minimal effort. For instance, it could adapt to a worker's unique style and background when creating content. Another example of this came from P1, a professor, who used ChatGPT to create starter drafts of a grant proposal. Instead of starting from scratch, P1 had ChatGPT analyse the introduction of a previous successful proposal written by P1. ChatGPT then used the same style, organisation, and tone to craft an introduction for a new grant proposal from a set of bullet points and incomplete thoughts. This approach allowed P1 to efficiently start the new proposal with an introduction that matched P1's personal writing style, which could then be refined and expanded upon to match the new grant.

As a higher-education instructor, P21 showcases how ChatGPT serves as an originator by revamping a class activity on the Business Model Canvas, a strategic management tool, through the creation of 20 new, more relevant case studies. Instead of focusing solely on quantity, P21 utilised ChatGPT to craft novel cases that better engage students in their 20s without having to spend extensive time generating these ideas from scratch.

The originator role also applies to more technical roles when real-world data can become too expensive and time-consuming to obtain. P5, a User Experience (UX) designer, discussed using generative AI models to create fake users that would provide responses to user research questions. P5 discussed the severe limitations of this practice, since most of the feedback from the 'user' (generative AI generated response) comes directly from what is mentioned in the prompt. Nonetheless, it is far cheaper and faster than scheduling actual user interviews, so it may provide some value when alternative user research methods are infeasible.

4.2.2. Brainstormer

Participants indicated that besides functioning as a springboard (often in the form of raw materials and

rudimentary foundations), it also served as a platform off which to iteratively bounce ideas. In other words, the system acts as creative catalysts in the brainstorming process, particularly for creative tasks. This enhances the creative process not by providing the initial ideas, as in the originator role, but also by facilitating back-and-forth interaction that enriches the participant's thinking process. This information practice leverages the system's input (in response to a prompt) as a foundation for further exploration.

P3, a specialist at a university's writing centre, demonstrated this iteration process. She would use ChatGPT to produce interview questions. She would prompt it to generate 10 questions, then ask for ChatGPT to generate more tailored questions depending on the needs of a specific position. For her, the generated questions were not necessarily the questions she would use during interviews, but they made her think about how she could 'transform' them and '*make [them] appropriate to [her] setting.*' ChatGPT provided raw material for her to adapt to her specific context and provided interaction that prompted her to ideate new interview questions.

As another example of iterative brainstorming with generative AI, P27, a biochemistry researcher, utilised ChatGPT to weigh the pros and cons of various experimental methods. This process not only enhanced P27's confidence in problem-solving, but also deepened her comprehension by providing a thorough analysis before and after experiments. Initially, ChatGPT's evaluations informed the experimental design, optimising the probabilities of success. Following unexpected results, the LLM's insights were crucial for revisiting the experiment's steps to identify any oversights. This iterative dialogue between the participant and ChatGPT proved invaluable, equipping the participant with a working knowledge base for discussions with mentors and peers about method selection and experimental outcomes. As a result, P27's confidence and depth of understanding improved from ChatGPT and, in turn, this improvement significantly elevated the quality of the subsequent academic exchanges.

4.3. Information evaluation

Information evaluation entails the examination and validation of information to ensure its accuracy and completeness. This information practice ensures that the information is reliable, factual, and error-free. This is often achieved through comparison with a known system of standards, such as the rules of English grammar and syntax.

4.3.1. Verifier

AI systems' contributions as verifiers lie in their ability to analyse and assess content for accuracy, relevance, and coherence. This role involves examining information, providing insights, and ensuring relevance and cohesion so that the information generated aligns with the context and requirements of the participant's query.

P22, a healthcare strategy consultant, used the verification capabilities of ChatGPT to indirectly determine whether certain healthcare data existed or, at least, was publicly available. P22 would ask ChatGPT to answer a question on something very specific in the healthcare sector (using it as a retriever). ChatGPT's response was either (1) the information P22 needed or (2) a vague or otherwise unconvincing answer that led P22 to infer that the information was unavailable (now turning into the verifier function). This typically meant that ChatGPT provided an answer that did not directly address the stated need in the prompt, and P22 subsequently inferred that the corpus of data needed to answer the question did not exist (or, at least, was inaccessible). This highlights that ChatGPT's role as a verifier necessitates trust between system and user. P22 trusted ChatGPT's ability to scrape the Internet so much so that an unsatisfactory answer from ChatGPT meant, to P22, that the information P22 did not exist, given that it did not seem to exist in ChatGPT's corpus.

P10, a solutions engineer, explained that there was also great value in ChatGPT's role as a 'rubber duck' in 'rubber duck programming,' a practice in software development where speaking to an inanimate object serves as self-reflexive verification for the logical processes being implemented. In this example, ChatGPT goes beyond the normal rubber duck since it can generate responses to the problem, but the premise remains of a communication outlet for the worker to verify and clarify their thought process without a human on the other end.

4.3.2. Reviewer

As a reviewer of information, the AI system's primary role lies in their proficiency in proofreading and editing text to ensure clarity and grammatical accuracy, thereby enhancing the overall quality and structure of written content. P13 highlighted this function of ChatGPT for proofreading texts, *'So I started using it, and it just helps me to proofread emails and reports. And sometimes my brain is tired after a whole day of meetings, and I have to write this last piece of writing.'*

P20, a solo entrepreneur for research tasks, used ChatGPT as an auxiliary 'wordsmith' for grant proposals and business emails. Specifically, ChatGPT helped her with diction suggestions for professional writing by

'saying things in ... the more formal way.' This reviewer function was particularly useful for ESL (English as a second language) students like P2, who initially signed up for ChatGPT to help proofread papers and 'polish' grammar.

P28, a digital media and program manager, mentioned that the structural recommendations were particularly helpful when writing a short story: *'It was actually telling me like, 'Oh, if you have more of an arc on a character, that is going to be more effective or if you display emotions.'* This also demonstrated generative AI systems' abilities for review extend beyond grammar and syntax into higher level characterisations regarding the effectiveness of a written message/story.

4.4. Information explanation

Information explanation and interpretation involve processes of clarifying, elucidating, and providing meaning to information or content (e.g. programming codes or complicated technical concepts). This practice focuses not only on simplifying complex information into accessible and comprehensible knowledge but also on interpreting and explaining the underlying significance or implications of that information. Therefore, the goal in this context is to expand on explanations and to unpack content rather than just to represent, synthesise, review or condense it. This practice makes information more accessible and relevant to specific contexts, considering different interpretations to enrich the user's overall comprehension.

4.4.1. Interpreter

As an interpreter, the system not only helps with translating and clarifying specific information, such as language or technical jargon, but also with facilitating a deeper understanding of the diverse interpretations of that information. The system can help users more effectively navigate through nuances, cultural contexts, and subjective viewpoints. Generative AI allows them to grasp not only the literal meaning but also potential interpretations and nuances. P12, a genetics researcher, noted how ChatGPT helped interpret key metrics in scientific experiments. When asked to explain the meaning of an MS (mass spectrometry) score in a particular kind of experiment, ChatGPT provided a theoretical description of what the MS score meant. P12 then followed up by asking about the meaning of a specific MS score, which ChatGPT then contextualised for the experiment.

As another example, P16 asked ChatGPT to describe *'what each of the lines [of code] are doing'* for old MATLAB files that had been progressively added to and passed down by fellow PhD students. For clarity,

P16 would copy and paste the lines of code into the ChatGPT prompt, and ChatGPT would return a response with the explanations embedded between the blocks of code. This example of ChatGPT writing pseudocode/comments was a particularly good instance of ChatGPT being able to turn what was merely data (no context or understanding on the part of the user) into actual information that was meaningful to the user.

4.4.2. Teacher

As a teacher or explainer, the system assists in imparting holistic understanding more broadly, which sets it apart from mere information synthesis or interpretation. This role of AI involves breaking down complex concepts, providing detailed explanations, and unpacking ideas. This aids in fostering learning and arriving at the big picture, as opposed to merely clarifying, translating, or interpreting specific pieces of information.

P21 highlighted a use case where ChatGPT adopted the teacher role and provided simplified explanations for queried topics. Specifically, while sitting in on a visiting faculty's academic job talk, P21 asked ChatGPT to explain an economic concept – 'causal inference' – that they had known a little about. ChatGPT then provided a refresher which helped ensure P21's general understanding of the topic was aligned with the bigger picture of that concept.

4.5. Information communication

The practice of information communication involves the effective transmission and sharing of information and ideas with other parties. It encompasses selecting the appropriate tone, crafting the message for clarity and impact, and ensuring that the information is received and understood as intended by the target audience.

4.5.1. Clarifier

This system's role focuses on clarifying information to effectively communicate with diverse audiences and determining the most suitable tone. For example, it assists users by analysing linguistic subtleties and contextual nuances and by providing recommendations to enhance message clarity, persuasiveness, and audience-specific tailoring.

Common use cases that surfaced include using ChatGPT to check for the tone of an email. P13 highlighted the use of ChatGPT for checking the tone of email communications, stating: *'Sometimes the way she [her assistant with 20-year gap] responds to my communications, I feel like she's being salty or she doesn't take feedback very well. But with time, I'm learning*

that it's about communication skills. And I use ChatGPT to check my tone, to check her tone, to also help me to find those nuances of human people communicating from different generations. And that has been fascinating because it has helped me... to recognize when I was being a little bit too direct and I needed to soften with words like, 'Can you kindly do this?'

In some instances, participants included additional contextual information about the target audience. ChatGPT would then revise the text to ensure that the language was more conducive to communicating with the intended audience. This can mitigate issues like cultural communication gaps, as was the case for P13, or intergenerational communication gaps, as was the case for P22. P26 also pointed out that this function was particularly valuable for those who speak English as a second language, as they could use generative AI to ensure their language is not only grammatically correct but also achieves the tone they seek in their writing. P3, a writing specialist who supports international students, echoed this point. Many of the students she had worked with asked ChatGPT for specific points on how to sound 'polite' or 'respectful' because *'one of the most difficult things to master [for ESL students] are the conventions of politeness.'*

The value of this clarifier role is apparent when someone is communicating with others outside of their domain. For instance, P20 had to write a marketing proposal when applying for a grant even though they were not very experienced with marketing language. After seeing ChatGPT's recommended text, the participant used ChatGPT's response nearly verbatim since, as they put it, *'That's close enough ... I don't know because I've not done this particular thing that I'm trying to get money for.'* This example particularly demonstrated the value of AI for communication purposes since it could quickly incorporate domain-specific language in the appropriate context even when the user was not a domain expert.

In brief, information communication shifts the focus to the audience and effectiveness of the current message for the given audience(s). Usually, this consists of modifying the tone of the message and goes beyond the main ideas being communicated to the audience.

4.6. Human roles

This information practice perspective underscores the fact that humans are not merely users of information or technology but active participants who, in collaboration with these tools, significantly shape the outcomes and consequences of AI. As a result, the practical function and augmentative role of generative AI systems do not

stem entirely from the inherent properties of the tools themselves but are shaped by how users interpret and interact with them. When users engage with these systems, they engage in particular practices to incorporate the system's information into their work. This includes setting up the system to provide the most relevant information, such as through prompt engineering, verifying the information by consulting additional sources for credibility, and other practices, like referencing. These actions may address gaps in the system or enhance the capabilities of AI, such as contextualisation (see Table 3).

Humans must still assess the quality of AI outputs, particularly their consistency, precision, and relevance. This critical analysis is particularly difficult due to generative AI's limited ability to explain its decision-making process when reaching conclusions. P26 believes this is a key barrier to widespread adoption of ChatGPT in certain settings since the logical reasoning that ChatGPT uses is not clear to those reviewing its response. Further, as P21 pointed out, the ability to critique responses hinges on one's pre-existing baseline knowledge of a topic.

These systems' ability to synthesise information often lacks backing and clear evidence from credible sources. Even though they did not necessarily view the information as false or misleading, several participants recognised the issue of data provenance, noting that the system might have struggled to support ideas, arguments, and claims with trustworthy sources (if any sources). For P3, they accepted this limitation for generative AI and described it like this, '*It's [generative AI] where I go for ideas, but not for information.*' P3 then goes on to personally verify the authenticity and credibility of any claims from generative AIs.

Table 3. Key roles played by humans in shaping the outcomes of generative AI.

Human Roles	Description
Critiquing	Assessing the quality, consistency, and relevance of AI outputs, and identifying the limitations, especially due to AI's lack of transparency in decision-making.
Integrating into Work	Incorporating AI-generated outputs into daily tasks and transforming them into valuable knowledge assets that align with work processes.
Contextualising	Placing AI outputs within broader organisational and work contexts to ensure relevance and alignment with goals.
Communicating to Stakeholders	Adapting AI-generated content to meet the diverse communication needs of various stakeholders, ensuring consistency and authenticity in tone and style.
Individualising	Personalising AI-generated content by adding unique elements like tone and style to maintain the distinct voice and engagement with the target audience.

A key aspect of this analysis is familiarity with the system, including its capabilities in various information practices and its limitations. Participants may have acquired this knowledge through public information about the system or their own experimentation over time. For example, P21 demonstrated an understanding of the system and its application areas by acknowledging that the system's training was limited to the knowledge available up until 2021 (for that specific version of ChatGPT).

4.6.1. Integrating into work processes

Workers aim to seamlessly integrate generative AI outputs into their daily tasks. This leads to the essential human role of 'integration work' in transforming AI responses into valuable knowledge assets. This integration often involves seemingly straightforward tasks, such as using an AI-recommended tool to create a presentation, as seen with P6, or, in the case of P12, using AI-generated ideas in experimental design. These integration tasks are crucial, representing the moment when humans take responsibility for embedding the AI's responses into outputs that can be integrated into often complex work processes. As such, integration work is not just about completing a task; it is more about ensuring that the AI's contributions are effectively merged with human processes to enhance the overall work quality.

4.6.2. Contextualising

AI's capacities are still largely task-centric and can miss the broader context as a result. Integrating AI into work practices requires workers to contextualise the output by placing the information processed or produced by AI systems within the broader contexts of their tasks and organisational goals. This process goes beyond merely adding a personal touch; it involves embedding the AI-generated information into the wider scope of work and objectives, as highlighted by P5. Knowledge workers play a critical role in this since they must decide which AI-produced content is relevant in decision-making and what should be disregarded to ensure the final product aligns with their work and organisational goals.

4.6.3. Communicating to different stakeholders

Generative AI systems, like ChatGPT, bring a novel aspect to technology with their capacity to linguistically emulate emotional intelligence and empathy. These systems demonstrate a basic level of understanding regarding communication tones and audience contexts, particularly in the transmission of information. Nonetheless, the complex nature of knowledge work, which often involves engaging with multiple stakeholders, demands that workers leverage their social intelligence.

Workers must skillfully manage the diverse needs and expectations of these stakeholders, a challenge that extends beyond the capabilities of current AI.

In this context, the customisation of AI outputs is crucial to align with specific communication styles and stakeholder expectations. For example, despite guiding ChatGPT to a desired tone, P22 must manually refine its output for consistency and authenticity, underscoring the importance of human intervention in adapting to different audiences. This refinement process ensures that AI-generated content not only meets its intended purpose but also aligns with the professional identity of individuals or organisations. In sum, this communication gap highlights a key discrepancy between AI's current capabilities and the complex demands of human contexts.

4.6.4. Individualising

Participants frequently adapt the content produced by systems like ChatGPT by adding their unique flair and tone to make the information more personalised and engaging. This involves not only refining the AI-generated output to reflect their distinct style but also ensuring the content maintains the unique 'voice' of the author or presenter. The objective is to counter the tendency of generative AI to homogenise language with an overly standardised style and structure, thereby obscuring the fact that the content originated from AI. This ensures the delivery resonates with the target audience, who typically prefer content that seems personally crafted by the presenter rather than generated by an AI system.

This customisation process extends beyond vocabulary adjustments to include formatting consistency and writing patterns, which can indicate AI involvement, especially as the response length increases. P22 highlights these aspects, noting how the differences become more apparent in longer AI-generated responses. Additionally, P13 emphasises the importance of incorporating a 'personal touch' and 'storytelling' elements in their writing. These strategies are vital for making the AI-generated output more relatable and engaging, seamlessly blending AI-produced material into human-centric communication practices.

5. Discussion

Generative AI enhances the key practices of managing information that are central to the tasks of knowledge workers in our study. These enhancements significantly add value to the suite of tools that knowledge workers already employ, including both general-purpose tools like search engines and specialised resources such as

proprietary databases. However, our research highlights particular ways these enhancements stand out in practice: these systems excel not just in grasping the structure of language, including vocabulary and grammar, but also in understanding the nuances of conversation and language context. Thanks to sophisticated natural language processing (NLP) capabilities, they are capable of producing responses and dialogues that feel human-like to many participants, thereby making complex information more accessible and engaging. They also excel in tailoring their responses in real-time based on the context of the inquiry and feedback from users. This leads to unprecedented opportunities in areas like information synthesis through leveraging the unique ability of these AI systems to integrate scattered pieces of information, spark new ideas, and provide individualised recommendations.

The actual impact of systems, such as generative AI systems, in practice materialises through the interaction of humans and machines (Leonardi 2011). Our study highlights the active role of knowledge workers; this participation often ensures that power remains with the worker, even as AI supports the information practices foundational to the decision-making process. This collaboration does not result in full automation; instead, workers see themselves as part of a human-AI partnership. While generative AI systems have automated some routine or less critical tasks, the primary advantages align with the Human-Centred AI (HCAI) vision, which seeks to amplify, augment, and enhance human capabilities, fostering self-efficacy, creativity, and responsibility (Shneiderman 2022). Our findings reveal that, through ongoing experimentation with generative AI, knowledge workers and AI systems achieve a balanced human-AI partnership.

This partnership involves a dynamic division of labour within the new information landscape, with generative AI providing its unique and, to some extent, unprecedented capabilities, yet knowledge workers remaining central. A key aspect of this dynamism is the ongoing interaction between systems and knowledge workers that renders the boundaries of their roles fluid and flexible. However, knowledge workers ensure their relevance through the aforementioned human roles and collaborate effectively with AI by developing and using AI literacies. The subsequent sections will explore these dynamics in further detail.

5.1. A dynamic division of work between humans and AI

In line with the HCAI framework (Vassilakopoulou et al. 2022), our study foregrounds a complex

interdependence between workers and machines within both organisational and societal contexts. These interactions underscore a dynamic division of labour that is characterised by a rich assemblage of collaboration, mutual enhancement, and boundary negotiation.

The integration of emotional intelligence and anthropomorphic features in generative AI systems shapes users' perceptions of their utility and fosters trust, essential for effective human-AI collaboration (Pataranutorn et al. 2023). Despite these advancements in mimicking humans, our research highlights a crucial gap in the 'last mile' of intellectual and conceptual tasks that generative AI cannot independently resolve. This includes critical areas like fact-checking, applying personalised stylistic preferences, and contextualising information – domains where the nuanced understanding and expertise of humans are irreplaceable. This suggests that, consistent with observations from Lee, Liang, and Yang (2022), the role and focus of some workers in producing information shift towards verifying and contextualising it. Furthermore, generative AI's inclination towards generalisations may diminish the precision and applicability of its outputs and emphasises the critical role of human oversight and customisation.

Our analysis also shows a significant blending of human and AI contributions across various information practices, specifically in information practices like information verification. This convergence does not lead to a clear-cut division but rather to a fluid, ongoing partnership within shared practices. Such a model indicates that both human insights and AI capabilities are intertwined and suggests a collaborative framework where the strengths of each side are harnessed to enhance overall performance and innovation. This division of labour is explained by Mollick (2023) through the metaphor of Cyborgs, representing a fusion of machine and human elements that are deeply integrated. Instead of merely delegating tasks, Cyborgs intertwine their efforts with AI, engaging in a dynamic back-and-forth exchange, as described in the following section on iteration.

While this dynamic division of labour between humans and generative AI can result in new productivity gains (Cambon et al. 2023), it is important to highlight the ethical implications of this interaction. Generative AI carries the risk of amplifying existing biases present in training data. In addition, overreliance on AI systems may also result in an erosion of critical human skills and expertise, what has been previously described in the research as automation bias and irony of automation (Bainbridge 1983; Parasuraman and Manzey 2010). Finally, accountable and transparent decision-making – or lack thereof – can be another

concern when it is hard to pinpoint decision sources or the output of work between humans and AI. Addressing these ethical concerns requires ongoing vigilance from knowledge workers as well as rigorous AI governance frameworks in organisations.

5.2. Continuous iteration

Building on the dynamic division of labour between humans and generative AI systems, a key aspect of this collaboration is the continuous iteration that characterises the interaction between knowledge workers and AI. This process is not a simple linear sequence of actions but a complex, iterative cycle that allows for an evolving understanding and refinement of tasks and outcomes. Our findings suggest at the heart of this process is a dialogic exchange where initial inputs from the workers often need refinement to meet the nuanced demands of specific information practices. For example, the dialogue between the worker and the generative AI system frequently starts with a broad prompt that may not capture the full specificity required. Through a series of adjustments and clarifications, the user hones in on the desired output by engaging in a back-and-forth with the AI that mirrors a conversation more than a command-response dynamic.

The iterative nature of human-AI interaction promotes a deeper engagement with the material by enabling knowledge workers to leverage the generative AI's capabilities for initial information generation and synthesis. In turn, human expertise is applied in refining, contextualising, and applying this information. This not only increases the effectiveness of the information practices but also ensures that the final outcomes are aligned with organisational goals and cultural values.

5.3. AI literacy

Mastering AI literacies is crucial when working alongside AI systems, as developing AI literacy empowers humans to effectively use and collaborate with AI (Fui-Hoon Nah et al. 2023). Our findings emphasise that AI literacy not only indicates that participants actively contribute to shaping the system's outcomes but also emerges from ongoing interaction with the system. Therefore, AI literacy has a key impact on the outcomes of system performance since its impacts are partly rooted in the inherent characteristics of this technology. For example, generative AI systems have the unique combination of available information through advanced natural language understanding and generation capabilities which can then be combined with users' understanding of their specific work domains

and immediate information needs. Specifically, crafting prompts that include contextual details about the worker makes complex inquiries more comprehensible to the system, enhancing its ability to provide relevant and accurate responses. Over time, as users interact more with the system, they develop a theory that links the specific functions and capabilities of generative AI systems to their information practices and is informed by both public sources and recurrent experimentation. This evolving understanding underscores the dynamic nature of AI literacy that is shaped by the interplay between system capabilities and user expertise.

The current reliance on users' baseline knowledge to effectively critique AI responses similarly highlights the need for users to develop a robust understanding of how generative AI systems generate or synthesise information. This AI literacy enables users to identify potential biases and inaccuracies in AI outputs to ensure that users can effectively validate and contextualise information. Since the AI systems may not be able to independently verify all information, users must foster skills in prompt engineering, critical evaluation, and integration of AI-generated content into their information practice.

6. Conclusion

Our research highlights the transformative potential of generative AI in knowledge work by highlighting the need for a more comprehensive and grounded understanding of human-AI collaboration. Knowledge workers leverage generative AI to augment their information practices and work to navigate a new landscape where AI acts as a partner rather than a replacement. This partnership fosters a dynamic division of labour, where AI supports tasks ranging from information synthesis to complex decision-making, and workers contribute their unique insights and contextual knowledge. Additionally, by adopting a micro-process analysis of knowledge workers' real-world engagements with generative AI tools in their individual work contexts, our study highlights a myriad of sociotechnical complexities that are difficult to capture and are part-and-parcel with what modern knowledge work actually looks like on a day-to-day basis. In particular, knowledge workers will continually need to collaborate with a large and increasingly diverse set of stakeholders (e.g. generational gaps between workers), rapidly pick up and apply new concepts to solve complex and novel problems (e.g. biology researchers needing to learn to code), communicate through a wide variety of digital mediums (e.g. email, formal reports, proposals) and data types (e.g. numeric, text-based), and much

more. Embedded within each workday, with the incorporation of an increasing number of digital and intelligent tools, the complexities of knowledge work is far greater than we typically give credit for in the existing literature. This perspective moves far beyond a primarily task-centric point of view since the social and relational aspects of digitally-mediated work life take up a significant amount of emotional and cognitive effort, and for some populations of workers, this effort may be greater than it is for others (e.g. participants whose English is a second language).

6.1. Contributions to research

This study makes key contributions to understanding the adoption of generative AI systems in knowledge work. By employing a mixed-method approach, including digital diary studies and interviews, the research provides grounded empirical insight into the day-to-day human – AI interaction, revealing how generative AI tools are incorporated into workflows, and adapted to situational information needs. The study's findings highlight the dynamic human-AI partnership, where AI systems augment human capabilities rather than replace them. This partnership could enhance both efficiency and creativity in knowledge work, aligning with the Human-Centred AI (HCAI) perspective that emphasises collaboration between humans and AI systems to achieve more effective and ethical organisational outcomes (Capel and Brereton 2023).

This research also contributes to the literature on information practices by highlighting how generative AI systems may transform or augment these practices. The study's emphasis on the active role of knowledge workers in shaping AI outputs through practices like critiquing, integrating, and contextualising provide a framework to capture the continued contribution of human expertise in the age of AI. What we specifically add to the current research on information practice is the recognition that emerging AI systems function not only as support infrastructure but as actors with some levels of agency. In partnership with humans, these systems perform information practices. Hence AI systems are key participants in information practice, and our findings advance the evolving and dynamic division of work between humans and intelligent machines in the context of modern knowledge work.

Finally, by documenting a human – AI partnership model (rather than a replacement model), our work counters prevailing narratives of AI-human competition and provides a more nuanced understanding of how roles and responsibilities shift in an AI-assisted

environment. It highlights that human agency, oversight, and creativity remain central even as certain tasks become automated. **Our findings** point to the evolving skill set that knowledge workers need in order to effectively leverage AI tools. We observed that – as part of their daily routines and interaction of generative AI – participants developed and exercised new literacies – such as prompt engineering (crafting effective inputs for AI), critical appraisal of AI outputs, data curation, and cross-validation of information. Successful integration of AI requires more than just deploying the system; it also depends on human workers' active role in *upskilling* and adapting their practices.

6.2. Implications for practice

To address the challenges identified in the use of generative AI systems, several design changes and enhancements can be made to improve the performance, reliability, and overall impact of generative AI to support information practices in the context of knowledge work.

6.2.1. Enhancing transparency and explainability

Generative AI systems should provide clearer rationales for their outputs, more effectively outlining how results were derived and what data sources have been used in producing them. Currently these systems lack metacognition (clear capabilities to recognise the limit of their knowledge) and therefore can act unpredictably and even erratically in high-stake decision-making situations (Griot et al. 2025). Such deficiencies may often result in offloading an extra cognitive load to users to constantly and vigilantly verify the AI-generated outputs. Improved transparency can help users more effectively trust, adopt, and critically evaluate AI-generated information (Lipton 2018).

6.2.2. Mitigating bias and ensuring fairness

Mitigating bias and focusing on fairness are also essential to improving the current performance of generative AI. Given that the training data of current AI systems are often restricted to limited sources of knowledge and may therefore reflect and reinforce specific worldview, ensuring that training data encompasses a wide range of sources and perspectives can reduce biases embedded in the systems (Henriksen, Creely, and Mehta 2022). Continually monitoring and updating these datasets to include underrepresented voices and contexts are other areas of algorithmic improvement. In addition, implementing algorithms that detect and flag potential biases in real-time can inform users about possible issues in AI-generated content. In this context,

integrating real-time fact-checking capabilities that cross-reference outputs with verified databases and reliable sources can ensure the validity of AI-generated information (Choi, Zhang, and Stvilia 2023). For example, introducing features that allow users to flag uncertain information and prompting the AI to provide verification or indicate the confidence level of its responses can further support the reliability of generative AI.

6.2.3. Customisation and contextualisation

In addition, the design and performance of generative AI tools should allow greater tailoring to each user's context and needs. Our findings make it clear that the users still need to perform a large share of contextualisation. Specifically, there is the need to develop more context-aware AI systems that incorporate user-provided background information to make outputs more relevant and useful to the unique work context of each user. For example, providing users with more ability to customise output style, tone, and level of detail will help align AI-generated content with individual preferences of knowledge workers, their unique information practices, and organisational norms and routines.

6.2.4. Promoting user training and AI literacy

While AI systems such as those described in this research are largely consumerised and are experimented with by each individual knowledge worker, organisations can still invest in training programmes to boost AI literacy among workers. This training could include comprehensive tutorials, best-practice guides, and examples of effective usage. In addition, communities of practice through which workers share experiences with tools, common challenges, and practical tips are other effective mechanisms to support the ongoing collective sensemaking and adaptation process. By improving workers' understanding of AI and their skill in interacting with these systems, organisations can ensure the technology truly augments work performance at both individual and organisational levels.

6.2.5. Ethical considerations and governance

Finally, despite individual workers being at the centre of the learning and experimentation process, implementing robust ethical guidelines and governance frameworks – commonly referred to as AI guardrails – ensures responsible use at both individual and organisational levels (Eom et al. 2024). Clear policies on data usage, user privacy, and security help prevent misuse and protect sensitive information. Regular audits and assessments help identify potential ethical issues

proactively (Papagiannidis et al. 2023). Such governance measures align AI deployment with broader organisational values and societal norms, ensuring accountability in how AI is integrated into work processes.

6.3. Study limitations

Despite its contributions, this study has several limitations. First, the reliance on users' baseline knowledge to critique AI responses reveals a potential gap in the AI system's ability to independently validate information. This dependency suggests a weakness in the AI's design that could impact its effectiveness across diverse user groups, especially those with varying levels of expertise. Consequently, the AI's utility may be compromised when users lack sufficient knowledge to identify inaccuracies or biases in AI-generated outputs.

Second, the study focuses on a specific set of generative AI tools, primarily large language models like ChatGPT. While these tools represent a significant portion of current generative AI applications, the findings may not be fully generalisable to other types of generative AI, such as those focused on image or video generation. In addition, we recognise that the study's cohort-based methodology, while effective in capturing detailed user experiences, may limit the generalisability of the findings to broader populations of knowledge workers or other professional contexts.

Lastly, we recognise that our investigations focused on the performance of these systems within a specific timeframe, so it inexorably comes with clear limitations in capturing their shaping and long lasting effects on information practices of knowledge workers. This is an important limitation since generative AI systems are dynamic and emergent technology, so their capabilities and limitations evolve over time as they interact with users and new features are integrated by the developers of these systems. The same applies to users' perceptions and practice as they interact with evolving systems over time. This remains an inherent limitation of our research methodology, which is primarily a snapshot approach – exploring the generative AI systems used within a specific timeframe. As a result, specifics of the role of generative AI systems for information practices presented in this study may change in the future, and this can serve as an impetus for future research.

6.4. Recommendations for further research

Future research can address the limitations identified in this study by exploring mechanisms to improve the AI's capability to autonomously or collaboratively validate

and verify information. Developing specific algorithms that mitigate biases in AI outputs and incorporating more robust validation mechanisms could significantly improve the effectiveness of generative AI systems across diverse user groups.

Further research is also needed to explore the impact of generative AI systems in other domains beyond text generation, such as image and video creation, to provide a more comprehensive understanding of how these technologies influence different forms of knowledge work. In addition, studies that include a more diverse range of participants, particularly those from different cultural and professional backgrounds, could offer valuable insights into how generative AI systems are perceived and utilised in other social and work contexts.

Finally, investigating the long-term effects of integrating generative AI systems into knowledge work, including potential changes in work routines, skill development, and job satisfaction, could be another direction for future research. This line of research could help organisations better understand the implications of adopting AI systems and guide the development of strategies that maximise the benefits of human-AI collaboration while addressing potential challenges.

Ethics approval statement

Ethical approval for the research reported in this manuscript was sought and obtained from the Office of Human Research Ethics at the University of North Carolina (UNC). The submission, bearing Reference ID 396528, has been reviewed and was approved by the UNC Institutional Review Board (IRB). For further details please contact: IRBIS@unc.edu or 919-966-3113.

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